Baseline Retrieval vs. Cross-Encoder Re-Ranking

A Two-Stage RAG Study on the Digital Personal Data Protection Act, 2023

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GitHub: https://github.com/Harshalak02/rag-baseline-vs-reranking-dpdp
Corpus: The Digital Personal Data Protection Act, 2023 (India) — https://www.meity.gov.in/static/uploads/2024/06/2bf1f0e9f04e6fb4f8fef35e82c42aa5.pdf

1 Setup & Environment

- **Document.** The Digital Personal Data Protection Act, 2023 (DPDP Act): a 21-page Indian statute of 44 sections and a penalty Schedule. Chosen because legal text separates a general rule from its exceptions by many clauses, relies on negation ("shall not", "no civil court"), and encodes role-specific limits — the conditions that stress a retriever.
- **Bi-Encoder.** `BAAI/bge-small-en-v1.5` (384-d), cosine similarity via inner product on L2-normalised vectors.
- **Vector store.** FAISS `IndexFlatIP` (exact search, so no recall is lost to the index).
- **Cross-Encoder.** `cross-encoder/ms-marco-MiniLM-L-6-v2`.
- **Generator (LLM).** `openai/gpt-oss-120b` via the Groq API, `temperature`= 0. The same prompt, model and decoding are used for both answers; the retrieved context is the only variable.
- **Hardware.** Kaggle GPU (CUDA); retrieval also runs on CPU, the GPU affects only wall-clock timing.

Preprocessing (structure-aware chunking). The PDF is parsed *section by section* with a sequence-guarded regex (section N is accepted only if it follows $N-1$, rejecting cross-references such as "section 8"). Each section is then split at its sub-section boundaries `(1)`, `(2)`, ..., and the penalty Schedule is re-paired into clean "breach → penalty" rows. Every chunk is labelled with its source, e.g. `[Section 9(3)]`. This deliberately places a rule and its exception in *separate* chunks (e.g. §9(1) consent vs. §9(4)/9(5) exemptions) — the failure mode this assignment targets. A self-test halts the run unless the sections are contiguous, the Schedule penalties survive, and the rule/exception separation holds. Final corpus: **149 chunks**.

Query set. 20 queries, chosen by evidence rather than hope: each candidate was screened by where its answer-bearing ("gold") chunk lands under the baseline, and the final set spans every outcome — *no-headroom* (gold already at rank 1–3), *rescue-possible* (ranks 4–10), and *recall-ceiling* (>10). Two controls verify the pipeline answers ordinary questions.

2 Two-Stage Retrieval

Both pipelines consume a *single* shared top-10 candidate set, so Pipeline 2's cost is exactly Pipeline 1's plus the re-rank step.

Pipeline 1 (Baseline). Embed the query, retrieve the top-10 by cosine similarity, pass the **top 3** to the LLM → *Answer 1.*

Pipeline 2 (Re-Ranking). Score the *same* 10 candidates with the Cross-Encoder, re-sort, pass the new **top 3** to the LLM → *Answer 2.*

3 Analysis

3.1 Architecture Comparison

A **Bi-Encoder** encodes query and passage *independently* into fixed vectors; relevance is their cosine similarity. Passage vectors never see the query, so they can be pre-computed offline and indexed, making search a fast nearest-neighbour lookup — but a single vector must compress the whole passage, blurring fine cues such as negation and "except when…". A **Cross-Encoder** concatenates query and passage (`[CLS] query [SEP] passage [SEP]`) and runs them through the transformer *together*, so self-attention operates jointly over all tokens and yields one relevance score. This captures token-level interactions far better, but nothing is cacheable: every (query, passage) pair needs its own forward pass. Hence the two-stage design — cheap high-recall bi-encoder retrieval, then expensive high-precision cross-encoder re-ranking on a short list.

3.2 Rank-Shift Analysis

Table 1 gives the rank of the gold chunk before and after re-ranking.

Table 1: Rank of the gold chunk, baseline vs. re-ranked (top-10 retrieved, top-3 used).

Query (abbrev.)	**Gold**	**Base**	**Re-rank**	**Effect**
Independent auditor & DPO in India	§10	1	1	both top-3
Relief for new small businesses (startups)	§17(3)	1	10	**demoted**
Use data without consent	§7	6	5	missed by both
Relaxed for research/archiving/statistics	§17(2)	3	1	both (sharpened)
Foreign customer data, Indian firm	§17(1)	3	4	demoted
Prohibited with children's data	§9	1	1	both top-3
What a person must NOT do	§15	4	1	**rescued**
Child data without parental consent	§9	1	1	both top-3
Fine for weak security safeguards	Sched.	1	1	both top-3
Max penalty for a Data Principal's breach	Sched.	2	1	both top-3
Sending data to other countries	§16	1	1	both top-3
Languages for the privacy notice	§5	1	1	both top-3
Nominate someone on death/incapacity	§14	1	1	both top-3
"Ask app to stop" → erase (synonym)	§8	1	2	both top-3
"Hires another firm" responsibility	§8	1	2	both top-3
Where to complain before the Board	§13	8	1	**rescued**
Board's powers (like a court?)	§28	5	2	**rescued**
Take a data dispute to a civil court	§39	7	1	**rescued**
Grounds to process data (control)	§4	1	1	both top-3
Who is a Data Principal (control)	§2	>10	>10	recall ceiling

Aggregate. Counting every query (a never-retrieved gold chunk is a miss, not dropped):

$$\text{Hit@3: } \underbrace{14/20 = 0.70}_{\text{baseline}} \rightarrow \underbrace{16/20 = 0.80}_{\text{re-ranked}}, \qquad \text{MRR: } 0.653 \rightarrow 0.753.$$

Re-ranking yields a **net +2** in Hit@3: **4 rescues** (§15, §13, §28, §39) against **2 demotions** (§17(3), §17(1)), with **1 recall ceiling** (§2) it cannot touch.

The four rescues — and why the baseline stumbled. Every rescued query shares a pattern: the operative clause uses vocabulary the query does *not*, so lexical/semantic proximity ranks *distractor* sections higher, and only the Cross-Encoder's joint attention recovers the true one.

- **§15 (#4→#1), the duties of a Data Principal.** Baseline top-3 were general consent/obligation chunks (§7, §4, §8) that overlap "giving her data" but lack the duties clause. *Answer 1:* "*Not found in the provided context.*" *Answer 2:* correctly lists the three prohibitions — not impersonating another person, not suppressing material information, not registering a false or frivolous complaint. A clear omission caused by the critical chunk sitting outside the top-3.
- **§13 (#8→#1), grievance before the Board.** The query ("where must I complain…") is dominated by the Board's complaint machinery (§27/§28), burying the clause that a Data Principal must "exhaust the opportunity of redressing her grievance" before approaching the Board.
- **§28 (#5→#2), the Board's investigative powers.** The answer — the Board has "the same powers as are vested in a civil court" — is phrased in civil-procedure language absent from the lay query.
- **§39 (#7→#1), bar of jurisdiction.** A pure negation: "*No civil court shall have the jurisdiction…*" — exactly the construction a bi-encoder's single vector blurs, and which joint attention resolves.

Re-ranking is not free (two demotions). For the startups query (§17(3)) the correct chunk fell **#1→#10**: the Cross-Encoder promoted *general* exemption chunks over the startup-specific one, and **Answer 2 regressed from a correct citation to "Not found"**. The foreign-outsourcing query (§17(1)) similarly slipped #3→#4. A re-ranker can *remove* a correct chunk, so its net value must be measured, not assumed.

Re-ranking cannot repair recall (a ceiling). For "Who is a Data Principal?" the definition chunk (§2) never entered the top-10, so the Cross-Encoder — which only re-orders the 10 candidates it is shown — could not surface it, and both answers returned "Not found". Re-ranking fixes *ordering*, not *recall*; widening `TOP_K` is the lever for recall.

3.3 Latency Profiling

Median of 5 repeats per query over 20 queries, warm-up discarded; corpus = 149 chunks on GPU.

Table 2: Retrieval latency per query (LLM generation runs separately on Groq's hardware).

Stage	Mean (ms)	Median (ms)	Std (ms)
Pipeline 1: baseline retrieval (embed query + FAISS top-10)	10.77	10.73	0.38
Cross-Encoder re-rank of 10 (query, chunk) pairs	25.77	26.24	3.41
Pipeline 2 total retrieval (baseline + re-rank)	**36.54**	**37.08**	3.44

Re-ranking adds ≈26 ms (∼3.4× baseline retrieval), far below typical LLM generation latency. Negligible for one query; the next section shows why it cannot be applied to the whole corpus.

3.4 Scaling: Why Not Cross-Encode the Entire Collection?

A Cross-Encoder cannot pre-compute anything — its score exists only for a specific (query, chunk) pair — so every query costs N forward passes with nothing cacheable. Measured directly,

scoring all 149 chunks took 0.31 s (≈2.1 ms/pair); extrapolating:

Table 3: Cross-encoding the full corpus at scale (per query, GPU).

Corpus size (chunks)	**Cross-Encoder time per query**
149 (this corpus)	0.31 s
10,000	≈0.3 min
100,000	≈3.5 min
1,000,000	≈34.6 min

By contrast the Bi-Encoder embeds the corpus *once* (1.7 s here) and answers each query in ≈11 ms, because a document's embedding is query-independent and can be indexed; with an ANN index (HNSW/IVF) search is sub-linear in N. Cross-encoding a million chunks would take tens of minutes *per query* — infeasible interactively. The two-stage design exists precisely to confine the cross-encoder's $O(N)$ cost to a fixed-size shortlist (here, 10 candidates).

4 Conclusion

Structure-aware (section/sub-section) chunking plus Schedule parsing gave the bi-encoder a strong first stage — both penalty questions were answered correctly (Rs. 250 crore and Rs. 10,000) once the Schedule table was re-paired. On a query set screened for difficulty, the Cross-Encoder delivered **four end-to-end rescues** (most strikingly §15 and §39, turning "Not found" into correct answers), lifting Hit@3 from 0.70 to 0.80 and MRR from 0.653 to 0.753. It is not a free win: two demotions and one untouchable recall ceiling show that a cross-encoder is a high-precision *re-ranker* whose benefit must be measured, and which can never replace first-stage retrieval because its cost grows linearly with the collection.